
Exercises Advanced Lesson 4

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 - Advanced Mathematical Modeling

Note: Some of the following exercises are expressed in every day mathematical notation rather than proper *Mathematica* notation. You MUST convert every day notation into *Mathematica* notation for your code to work properly. For example:

Equations must be changed from "=" to "==" to work correctly (i.e. $x^2 == 2 x$ not $x^2 = 2 x$).

Functions must be expressed as $f[x]$ and not $f(x)$ to work correctly.

Functions must be defined using "_" to work correctly (i.e. $f[x_] = 2 x^2 + 1$ and not $f[x] = 2 x^2 + 1$).

Mathematica functions must start with a capital (i.e. $\text{Sin}[x]$ not $\text{sin}[x]$).

Your functions should start with a lower case letter (i.e. $f[x]$ not $F[x]$).

Special symbols must be entered correctly (i.e. e from the palette and not e).

These exercises begin your training in making this conversion automatically.

Remember to clear all variables that you will use below: `Clear[ ... ]`

`Clear[a, b, f, g, x, y, u, v, someData];`

1: Using the existing code in *Mathematica*'s Help Documentation Center can make your work much easier.

Part1: Open *Mathematica*'s Help Documentation Center.

Part2: Look up ParametricPlot.

Part3: Open each and every example and see what all of the examples look like.

Part4: Under Option>> MeshShading, choose your favorite example of MeshShading and Copy and Paste the corresponding *Mathematica* Code into your notebook and then evaluate this code to see the plot appear in your notebook.

Part5: Make a second copy of the same code in your notebook just below your first copy. Then change the parametric interval from $0 \leq u \leq 4\pi$ to $2\pi \leq u \leq 8\pi$ and reevaluate to see the difference that it makes.

```
ParametricPlot[{u Cos[u], u Sin[u]}, {u, 0, 4  $\pi$ }, Mesh  $\rightarrow$  20, MeshShading  $\rightarrow$  {Red, Blue}]
```

```
ParametricPlot[{u Cos[u], u Sin[u]}, {u, 2  $\pi$ , 8  $\pi$ }, Mesh  $\rightarrow$  20, MeshShading  $\rightarrow$  {Red, Blue}]
```

2: Generate a graph that simultaneously contains the trigonometric functions: $\sin(x)$, $\cos(x)$, $\tan(x)$, $\cot(x)$ for $0 \leq x \leq 2\pi$.

```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }]
```

3: Extend the above graph to restrict the y-axis range to be: $-3 \leq y \leq 3$.

```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }, PlotRange  $\rightarrow$  {-3, 3}]
```

4: Extend the above graph so that each successive curve has the following thickness: Thickness [.01], Thickness [.005], Thickness [.003], Thickness [.001].

```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }, PlotRange  $\rightarrow$  {-3, 3},  
PlotStyle  $\rightarrow$  {Thickness[0.01`], Thickness[0.005`], Thickness[0.003`], Thickness[0.001`]}]
```

5: Extend the above graph so that the plot is labeled with: $\sin(x)$, $\cos(x)$, $\tan(x)$, $\cot(x)$.

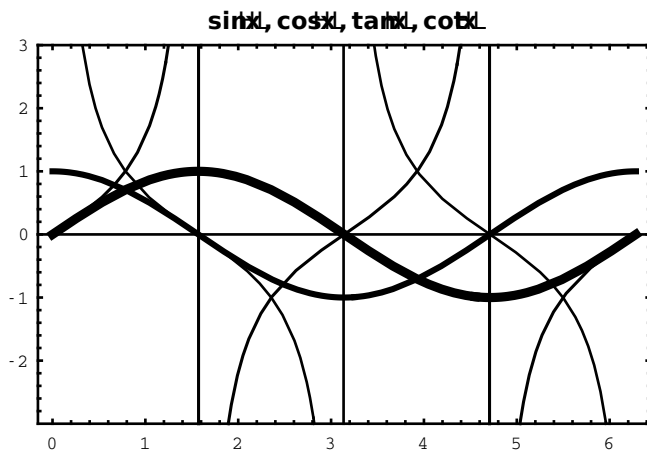
```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }, PlotRange  $\rightarrow$  {-3, 3},  
PlotStyle  $\rightarrow$  {Thickness[0.01`], Thickness[0.005`], Thickness[0.003`], Thickness[0.001`]},  
PlotLabel  $\rightarrow$  "sin(x), cos(x), tan(x), cot(x)"]
```

6: Extend the above graph so that the plot label is displayed in 12 point Arial-Bold font.

```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }, PlotRange  $\rightarrow$  {-3, 3},  
PlotStyle  $\rightarrow$  {Thickness[0.01`], Thickness[0.005`], Thickness[0.003`], Thickness[0.001`]},  
PlotLabel  $\rightarrow$  Style["sin(x), cos(x), tan(x), cot(x)",  
FontFamily  $\rightarrow$  "Arial", FontWeight  $\rightarrow$  Bold, FontSize  $\rightarrow$  12]]
```

7: Extend the above graph so that the graph is surrounded with a frame.

When you are through, your graph should look like:



```
Plot[{Sin[x], Cos[x], Tan[x], Cot[x]}, {x, 0, 2  $\pi$ }, PlotRange  $\rightarrow$  {-3, 3},
  PlotStyle  $\rightarrow$  {Thickness[0.01`], Thickness[0.005`], Thickness[0.003`], Thickness[0.001`]},
  PlotLabel  $\rightarrow$  Style["sin(x), cos(x), tan(x), cot(x)",
    FontFamily  $\rightarrow$  "Arial", FontWeight  $\rightarrow$  Bold, FontSize  $\rightarrow$  12], Frame  $\rightarrow$  True]
```

8: Using the data generated below:

i.e. `someData = Table[{0.1*i, (0.1*i)2 + 0.5*(RandomReal[] - 0.5)}, {i, 1, 20}]`

```
someData = Table[{0.1 i, (0.1 i)2 + 0.5 (RandomReal[] - 0.5)}, {i, 1, 20}]
```

Part1: Plot the list of data with points having a point size of 2% (i.e. .02) of the overall graph size.

Part2: Define a function $f(x)$ to be the function x^2 .

Plot the function $f(x)$ for $0 \leq x \leq 2$, where the curve has a thickness of 1% (i.e. .01) of the overall graph size.

Part3: Use the Show[...] command to display both of the above graphs at the same time.

```
SeedRandom[20 250 825];
someData = Table[{0.1` i, (0.1` i)2 + 0.5` (RandomReal[] - 0.5`)}, {i, 1, 20}]
plotData = ListPlot[someData, PlotStyle  $\rightarrow$  PointSize[0.02`]]

Clear[f];
f[x_] = x2;
plotF = Plot[f[x], {x, 0, 2}, PlotStyle  $\rightarrow$  Thickness[0.01`]]

Show[plotData, plotF]
```

9: Find the least-squares fit "g" to the data: someData generated in problem 8 using the functions: **{1, e^x}**.

```
Clear[g, x];
g = Fit[someData, {1, ex}, x]
```

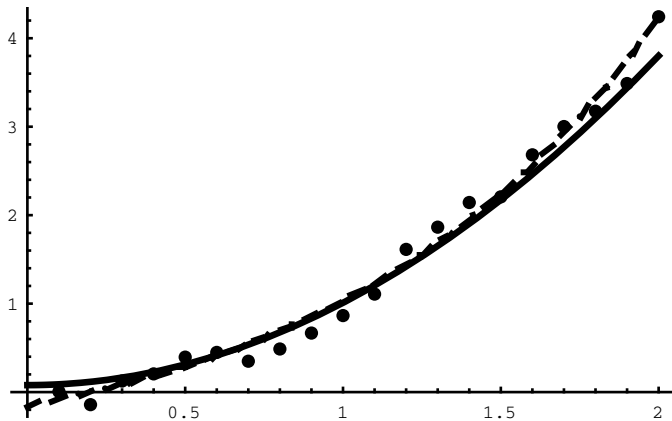
10: Plot the least-squares exponential fit "g" to the data so that it has Thickness [.009] and Dashing [{.02, .02}].

```
plotG = Plot[g, {x, 0, 2}, PlotStyle  $\rightarrow$  {Thickness[0.009`], Dashing[{0.02`, 0.02`}]}
```

11: Show the list plot of the data, the continuous plot of the function "f" and the continuous plot of the

function "g" on one graph.

When you are through, your graph should look like:



`Show[plotData, plotF, plotG]`

12: Plot the parametric equation: $\{\cos(7\theta)\cos(11\theta), \cos(7\theta)\sin(11\theta)\}$ for $0 \leq \theta \leq 2\pi$ so that it has Thickness 1% of the overall graph size and an aspect ratio of 1.

```
ParametricPlot[{Cos[7 t] Cos[11 t], Cos[7 t] Sin[11 t]},
  {t, 0, 2 π}, PlotStyle → Thickness[0.01], AspectRatio → 1]
```

13: Many times it is necessary to use several graphic representations to determine the behavior under analysis. This is especially true if the behavior to be observed is highly complex. For example, say that you want to understand the behavior of the function: $\sin(x / \sin(y))$ for: $-\pi \leq x \leq \pi$, $-\pi \leq y \leq \pi$.

Part 1: As a first attempt, generate a 3D plot of this function. (Why might there be problems generating this graph when $y = 0$ or for $y = \pi$ etc.?)

Part 2: Generate a 3D plot of this function where the PlotPoints are set to 20. (More resolution is needed.)

Part 3: Generate a 3D plot of this function where the PlotPoints are set to 50 and the Mesh to False. (More resolution is needed.)

Part 4: Generate a 3D plot of this function where the PlotPoints are set to 200 and the Mesh set to False. (Eventually, either the computational time will become excessive or the computer will prove to be inadequate.)

Part 5: Generate a density plot of this function with the Mesh set to False. (Why might there be problems generating this graph?)

Part 6: Repeat Part 2 for the density plot.

Part 7: Repeat Part 3 for the density plot.

Part 8: Repeat Part 4 for the density plot.

Part 9: Looking at the above 3D plots and the density plots, describe (in a Text Cell) the behavior of the

function $\sin(x / \sin(y))$ for: $-\pi \leq x \leq \pi$, $-\pi \leq y \leq \pi$.

```
Plot3D[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }]
```

```
Plot3D[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  20]
```

```
Plot3D[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  50, Mesh  $\rightarrow$  False]
```

```
Plot3D[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  200, Mesh  $\rightarrow$  False]
```

```
DensityPlot[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, Mesh  $\rightarrow$  False]
```

```
DensityPlot[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  20, Mesh  $\rightarrow$  False]
```

```
DensityPlot[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  50, Mesh  $\rightarrow$  False]
```

```
DensityPlot[Sin[ $\frac{x}{\sin[y]}$ ], {x, - $\pi$ ,  $\pi$ }, {y, - $\pi$ ,  $\pi$ }, PlotPoints  $\rightarrow$  200, Mesh  $\rightarrow$  False]
```

Part 1: $\sin(x/\sin y)$ blows up wherever $\sin y = 0$, which is at $y = 0$ and $y = \pm\pi$. Near those lines the function oscillates infinitely fast and the default sampling cannot keep up, so the surface looks torn.

Part 9: away from those lines the surface is a smooth ripple; as y approaches 0 or $\pm\pi$ the ripples crowd together without limit. Raising PlotPoints shows more of them but never resolves them, because there is no finest scale to reach. The density plots show the same thing as bands packing tighter toward the same lines.

14: Using Manipulate[...]:

Part 1: Create a dynamic 2-D plot of:

$$e^{-ax} \text{ where } 0 \leq x \leq 1$$

With a continuous slider for a where $0 \leq a \leq 10$

Part 2: Note that the plot jumps around as you move the slider. Add a PlotRange option to your plot (that plots All of the range) to fix this problem.

Part 3: Create a dynamic 3-D plot of:

$$e^{-ax-by} \text{ where } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1$$

With a continuous slider for a where $0 \leq a \leq 10$

With a discrete slider for b where $0 \leq b \leq 10$ in steps of 2

```
Manipulate[Plot[E^(-a x), {x, 0, 1}], {a, 0, 10}]
```

```
Manipulate[Plot[E^(-a x), {x, 0, 1}, PlotRange -> All], {a, 0, 10}]
```

```
Manipulate[
  Plot3D[E^(-a x - b y), {x, 0, 1}, {y, 0, 1}, PlotRange -> All],
  {a, 0, 10}, {b, 0, 10, 2}]
```

15: Using Manipulate[...]:

Create a dynamic algebraic calculator that expands (i.e. Expand[...]) the expression:

$$(x + y)^n$$

With a discrete slider for n where $0 \leq n \leq 10$ in steps of 1

```
Manipulate[Expand[(x + y)^n], {n, 0, 10, 1}]
```

16. Analyze the behavior of the equation:

$$2x^2 + 3x^3y = 4x^4 + 5y + 6$$

Part 1: Solve (i.e. use Solve[...]) the above equation for y in terms of x .

Part 2: Use the results from Part 1 to define a function $f(x)$ where $y = f(x)$.

Remember, you have an expression for y in terms of x from Part 1.

Part 3: Plot $f(x)$ for $-4 \leq x \leq 4$.

Part 4: Calculate $f'(x)$.

Part 5: Plot $f'(x)$ for $-4 \leq x \leq 4$.

Part 6: Find the critical points for $f(x)$ by solving (i.e. use Solve[...]) the equation

$$f'(x) = 0$$

for x . Note that the result is not very useful in this form (i.e. the answer contains Root[...] objects).

Part 7: Find the critical points for $f(x)$ numerically by solving (i.e. use NSolve[...]) the equation

$$f'(x) = 0$$

for x . Note that the result has two complex and four real values.

Part 8: Comment (in a Text Cell) on whether or not the four real roots found in Part 7 agree with the plot in Part 5. Also, which of the four real roots correspond to local maxima of $f(x)$?

```
Clear[f, x, y];
sol16 = Solve[2 x^2 + 3 x^3 y == 4 x^4 + 5 y + 6, y]
f[x_] = y /. First[sol16]
Plot[f[x], {x, -4, 4}]
Simplify[f'[x]]
Plot[f'[x], {x, -4, 4}]
Solve[f'[x] == 0, x]
NSolve[f'[x] == 0, x]
roots16 = Chop[x /. NSolve[f'[x] == 0, x]]
realRoots16 = Select[roots16, Im[#1] == 0 &]
TableForm[Table[{r, Chop[f'[r]], Chop[f''[r]], If[f''[r] < 0, "local maximum", "local minimum"]},
  {r, realRoots16}], TableHeadings -> {None, {"x", "f'[x]", "f''[x]", "classification"}}]
```

Part 8: NSolve gives six roots, two complex and four real, at about $x = -0.84, 0, 0.27$ and 1.96 . Those are exactly the four places where the plot of $f'(x)$ crosses zero, so they agree. Checking the second derivative there, $x = -0.84$ and $x = 0.27$ are the local maxima (f'' negative), while $x = 0$ and $x = 1.96$ are local minima.